MACHINE LEARNING LAB – OBSERVATION RECORD

EXPERIMENT 1: FIND-S ALGORITHM

Experiment No.: 1

Date: _14/10/2026_____

Experiment Question: Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

FIND-S: General concept and mathematical rule

FIND-S is a concept-learning algorithm. It begins with the most specific hypothesis and processes only positive training examples. For each attribute, it keeps the value if all positive examples seen so far agree; otherwise, it replaces the attribute with "?". The symbol "?" means any value is accepted. The symbol "Ø" below represents an attribute that has not yet been assigned.

Hypothesis representation: $H = (h_1, h_2, \ldots, h_n)$

For a positive example $x = (x_1, x_2, \ldots, x_n)$, update each attribute as:

$h_i(new) = x_i$, if $h_i = Ø$; $h_i(new) = h_i$, if $h_i = x_i$; $h_i(new) = ?$, if $h_i \neq x_i$.

Negative examples are ignored. FIND-S does not use negative examples to specialize its hypothesis.

Important: the final hypothesis summarizes these training examples; it does not prove a real-world cause-and-effect relationship.

Subproblem 1: Enjoy Sport Prediction

1. Aim

To implement FIND-S and learn the most specific hypothesis for predicting whether a person enjoys a sport from weather conditions.

2. Concept

The attributes are Sky, Temperature, Humidity, and Wind. The target is Enjoy Sport. Only rows labelled Yes are used to update the hypothesis.

3. Dataset Description

The training examples are shown below. The last column is the target class.

Sky	Temperature	Humidity	Wind	Enjoy Sport
Sunny	Warm	Normal	Strong	Yes
Sunny	Warm	High	Strong	Yes
Rainy	Cold	High	Strong	No
Sunny	Warm	High	Weak	Yes

4. Algorithm

1. Initialize H to the most specific hypothesis (Ø in every attribute).
2. Read each training example from left to right.
3. If the target is No, ignore the example.
4. If the target is Yes, compare each attribute with H.
5. Replace Ø with the example value; if a value conflicts with the current specific value, replace it with ?; otherwise keep it unchanged.
6. Display the final hypothesis after all examples have been processed.

5. Step-by-step Procedure and Detailed Calculations

7. Start: $H_0 = (Ø, Ø, Ø, Ø)$.
8. Example 1 is positive: (Sunny, Warm, Normal, Strong). Replace each Ø with the corresponding value. H_1 = (Sunny, Warm, Normal, Strong).
9. Example 2 is positive: (Sunny, Warm, High, Strong). Sky, Temperature, and Wind match. Humidity differs (Normal ≠ High), so generalize that attribute. H_2 = (Sunny, Warm, ?, Strong).
10. Example 3 is negative (No), so FIND-S ignores it. H_3 = (Sunny, Warm, ?, Strong).
11. Example 4 is positive: (Sunny, Warm, High, Weak). Sky and Temperature match; Humidity is already ?; Wind differs (Strong ≠ Weak), so generalize Wind. H_4 = (Sunny, Warm, ?, ?).

6. Formulae

Let $H = (h_1, h_2, \ldots, h_n)$ be the current hypothesis and $x = (x_1, x_2, \ldots, x_n)$ be a positive training example.

Attribute update:

• If $h_i = Ø$, then $h_i(\text{new}) = x_i$.

• If $h_i = x_i$, then $h_i(\text{new}) = h_i$ (no change).

• If $h_i \neq x_i$ and $h_i \neq Ø$, then $h_i(\text{new}) = ?$.

Final hypothesis for this dataset: Hfinal = (Sunny, Warm, ?, ?).

No numerical arithmetic is required for FIND-S: the “calculation” is a stepwise comparison of categorical attribute values. A mismatch causes generalization to ?.

7. Python Program

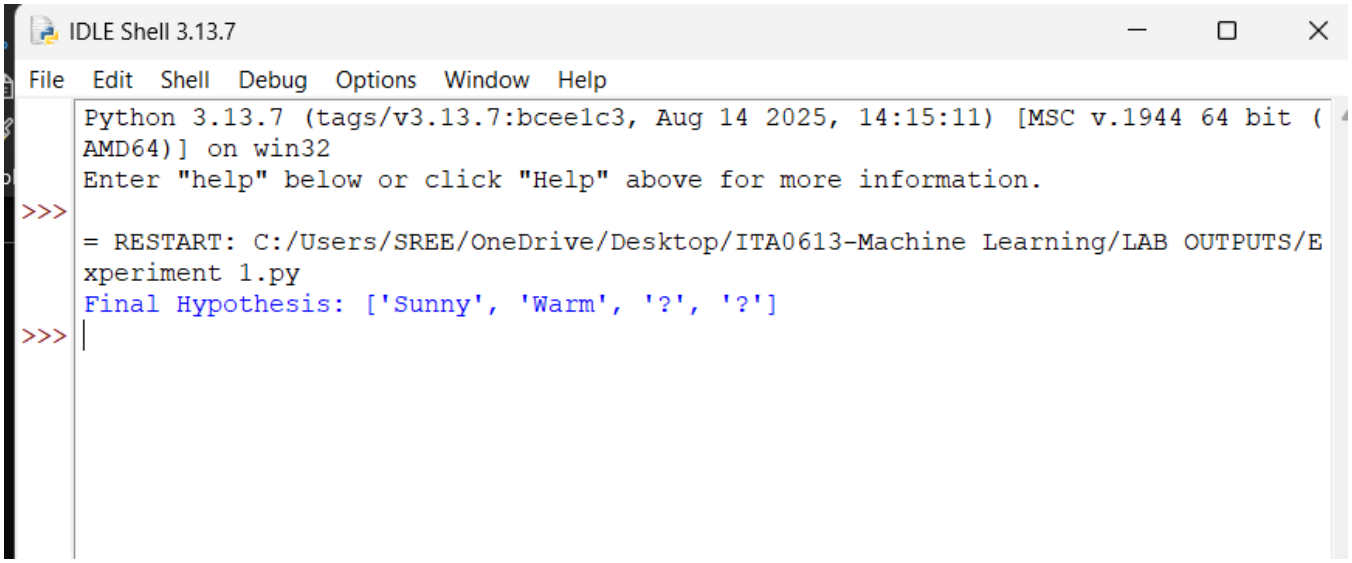
```
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'No'],
    ['Sunny', 'Warm', 'High', 'Weak', 'Yes']
]

h = ['Ø'] * 4

for row in data:
    if row[-1] == 'Yes':
        for i in range(4):
            if h[i] == 'Ø':
                h[i] = row[i]
            elif h[i] != row[i]:
                h[i] = '?'

print("Final Hypothesis:", h)
```

8. Output



9. Result / Conclusion

The FIND-S algorithm was implemented successfully for this dataset. Final hypothesis: (Sunny, Warm, ?, ?). The learned hypothesis requires Sky = Sunny and Temperature = Warm; any Humidity and Wind values are allowed by this hypothesis.

Subproblem 2: Weather Suitability Prediction

1. Aim

To apply FIND-S to learn the common conditions in examples labelled Suitable Weather = Yes.

2. Concept

The attributes are Sky, Temperature, Humidity, and Wind. The positive examples are used to generalize the hypothesis; the negative example is ignored.

3. Dataset Description

The training examples are shown below. The last column is the target class.

Sky	Temperature	Humidity	Wind	Suitable Weather
Sunny	Hot	High	Weak	Yes
Sunny	Hot	High	Strong	Yes
Rainy	Cold	High	Weak	No
Sunny	Mild	Normal	Weak	Yes

4. Algorithm

12. Initialize H to the most specific hypothesis (Ø in every attribute).
13. Read each training example from left to right.
14. If the target is No, ignore the example.
15. If the target is Yes, compare each attribute with H.
16. Replace Ø with the example value; if a value conflicts with the current specific value, replace it with ?; otherwise keep it unchanged.
17. Display the final hypothesis after all examples have been processed.

5. Step-by-step Procedure and Detailed Calculations

18. Start: H_0 = (Ø, Ø, Ø, Ø).
19. Example 1 is positive: (Sunny, Hot, High, Weak). H_1 = (Sunny, Hot, High, Weak).
20. Example 2 is positive: (Sunny, Hot, High, Strong). Wind differs (Weak ≠ Strong), so H_2 = (Sunny, Hot, High, ?).
21. Example 3 is negative (No), so it is ignored. H_3 = (Sunny, Hot, High, ?).
22. Example 4 is positive: (Sunny, Mild, Normal, Weak). Temperature differs (Hot ≠ Mild), so Temperature becomes ?. Humidity differs (High ≠ Normal), so Humidity becomes ?. Sky remains Sunny and Wind remains ?. H_4 = (Sunny, ?, ?, ?).

6. Formulae

Let $H = (h_1, h_2, \ldots, h_n)$ be the current hypothesis and $x = (x_1, x_2, \ldots, x_n)$ be a positive training example.

Attribute update:

- If h_i = Ø, then $h_i(new) = x_i$.
- If $h_i = x_i$, then $h_i(new) = h_i$ (no change).
- If $h_i \neq x_i$ and $h_i \neq$ Ø, then $h_i(new)$ = ?.

Final hypothesis for this dataset: Hfinal = (Sunny, ?, ?, ?).

No numerical arithmetic is required for FIND-S: the “calculation” is a stepwise comparison of categorical attribute values. A mismatch causes generalization to ?.

7. Python Program

```
data = [
    ['Sunny', 'Hot', 'High', 'Weak', 'Yes'],
    ['Sunny', 'Hot', 'High', 'Strong', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Weak', 'No'],
    ['Sunny', 'Mild', 'Normal', 'Weak', 'Yes']
]

h = ['Ø'] * 4

for row in data:
    if row[-1] == 'Yes':
        for i in range(4):
            if h[i] == 'Ø':
                h[i] = row[i]
            elif h[i] != row[i]:
                h[i] = '?'

print("Final Hypothesis:", h)
```

8. Output

```
>>>
= RESTART: C:/Users/SREE/OneDrive/Desktop/ITA0613-Machine Learning/LAB OUTPUTS/E
xperiment 1.py
Final Hypothesis: ['Sunny', '?', '?', '?']
>>>
```

9. Result / Conclusion

The FIND-S algorithm was implemented successfully for this dataset. Final hypothesis: (Sunny, ?, ?, ?). Among the positive examples, Sky remains Sunny while Temperature, Humidity, and Wind vary.

Subproblem 3: Student Performance Prediction

1. Aim

To use FIND-S to identify the attribute pattern shared by students whose result is Pass.

2. Concept

The attributes are Attendance, Internal Marks, and Assignment Submission. The target is Result. In the program, Yes represents Pass and No represents Fail.

3. Dataset Description

The training examples are shown below. The last column is the target class.

Attendance	Internal Marks	Assignment	Result
High	Good	Regular	Yes
High	Good	Irregular	Yes
Low	Poor	Regular	No
Medium	Good	Regular	Yes

4. Algorithm

23. Initialize H to the most specific hypothesis (Ø in every attribute).
24. Read each training example from left to right.
25. If the target is No, ignore the example.
26. If the target is Yes, compare each attribute with H.
27. Replace Ø with the example value; if a value conflicts with the current specific value, replace it with ?; otherwise keep it unchanged.
28. Display the final hypothesis after all examples have been processed.

5. Step-by-step Procedure and Detailed Calculations

29. Start: H_0 = (Ø, Ø, Ø).
30. Example 1 is positive: (High, Good, Regular). H_1 = (High, Good, Regular).
31. Example 2 is positive: (High, Good, Irregular). Assignment differs (Regular ≠ Irregular), so H_2 = (High, Good, ?).
32. Example 3 is negative (No), so it is ignored. H_3 = (High, Good, ?).
33. Example 4 is positive: (Medium, Good, Regular). Attendance differs (High ≠ Medium), so Attendance becomes ?. Internal Marks remains Good and Assignment is already ?. H_4 = (?, Good, ?).

6. Formulae

Let H = (h_1, h_2, …, h_n) be the current hypothesis and x = (x_1, x_2, …, x_n) be a positive training example.

Attribute update:

• If h_i = Ø, then h_i(new) = x_i.

• If h_i = x_i, then h_i(new) = h_i (no change).

• If h_i ≠ x_i and h_i ≠ Ø, then h_i(new) = ?.

Final hypothesis for this dataset: Hfinal = (?, Good, ?).

No numerical arithmetic is required for FIND-S: the “calculation” is a stepwise comparison of categorical attribute values. A mismatch causes generalization to ?.

7. Python Program

```
data = [
    ['High', 'Good', 'Regular', 'Yes'],
```

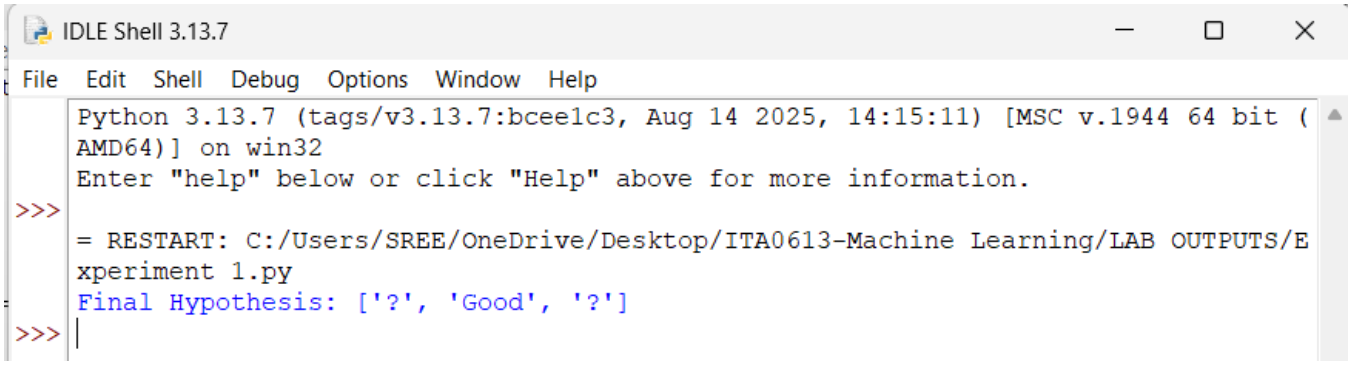
```
    ['High', 'Good', 'Irregular', 'Yes'],
    ['Low', 'Poor', 'Regular', 'No'],
    ['Medium', 'Good', 'Regular', 'Yes']
]

h = ['Ø'] * 3

for row in data:
    if row[-1] == 'Yes':
        for i in range(3):
            if h[i] == 'Ø':
                h[i] = row[i]
            elif h[i] != row[i]:
                h[i] = '?'

print("Final Hypothesis:", h)
```

8. Output



9. Result / Conclusion

The FIND-S algorithm was implemented successfully for this dataset. Final hypothesis: (?, Good, ?). The positive examples share Good internal marks; attendance and assignment status vary in this small dataset.

Subproblem 4: Animal Classification

1. Aim

To implement FIND-S to learn the common attribute pattern of animals belonging to a target category.

2. Concept

The attributes are Body Temperature, Number of Legs, and Habitat. The target label is Animal Category (Yes/No).

3. Dataset Description

The training examples are shown below. The last column is the target class.

Body Temperature	Legs	Habitat	Animal Category
Warm	Four	Land	Yes
Warm	Four	Land	Yes
Cold	Two	Water	No
Warm	Four	Water	Yes

4. Algorithm

34. Initialize H to the most specific hypothesis (Ø in every attribute).
35. Read each training example from left to right.
36. If the target is No, ignore the example.
37. If the target is Yes, compare each attribute with H.
38. Replace Ø with the example value; if a value conflicts with the current specific value, replace it with ?; otherwise keep it unchanged.
39. Display the final hypothesis after all examples have been processed.

5. Step-by-step Procedure and Detailed Calculations

40. Start: $H_0 = (Ø, Ø, Ø)$.
41. Example 1 is positive: (Warm, Four, Land). H_1 = (Warm, Four, Land).
42. Example 2 is positive and identical to Example 1, so no change occurs. H_2 = (Warm, Four, Land).
43. Example 3 is negative (No), so it is ignored. H_3 = (Warm, Four, Land).
44. Example 4 is positive: (Warm, Four, Water). Habitat differs (Land ≠ Water), so Habitat becomes ?. H_4 = (Warm, Four, ?).

6. Formulae

Let $H = (h_1, h_2, \ldots, h_n)$ be the current hypothesis and $x = (x_1, x_2, \ldots, x_n)$ be a positive training example.

Attribute update:

- If $h_i = Ø$, then $h_i(new) = x_i$.
- If $h_i = x_i$, then $h_i(new) = h_i$ (no change).
- If $h_i \neq x_i$ and $h_i \neq Ø$, then $h_i(new) = ?$.

Final hypothesis for this dataset: Hfinal = (Warm, Four, ?).

No numerical arithmetic is required for FIND-S: the "calculation" is a stepwise comparison of categorical attribute values. A mismatch causes generalization to ?.

7. Python Program

```
data = [
    ['Warm', 'Four', 'Land', 'Yes'],
    ['Warm', 'Four', 'Land', 'Yes'],
```

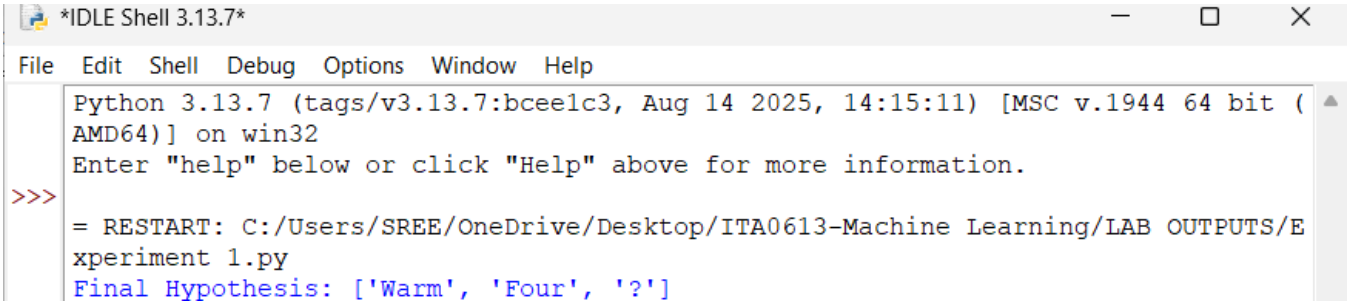
```
    ['Cold', 'Two', 'Water', 'No'],
    ['Warm', 'Four', 'Water', 'Yes']
]

h = ['Ø'] * 3

for row in data:
    if row[-1] == 'Yes':
        for i in range(3):
            if h[i] == 'Ø':
                h[i] = row[i]
            elif h[i] != row[i]:
                h[i] = '?'

print("Final Hypothesis:", h)
```

8. Output



9. Result / Conclusion

The FIND-S algorithm was implemented successfully for this dataset. Final hypothesis: (Warm, Four, ?). The positive examples share Warm body temperature and Four legs, while Habitat varies.

Subproblem 5: Computer Purchase Prediction

1. Aim

To apply FIND-S to learn the common attributes of computers labelled as purchased.

2. Concept

The attributes are Price, Speed, and Memory. The target is Purchase (Yes/No). Positive examples update the hypothesis.

3. Dataset Description

The training examples are shown below. The last column is the target class.

Price	Speed	Memory	Purchase
High	Fast	Large	Yes
High	Fast	Small	Yes
Low	Slow	Small	No
Medium	Fast	Large	Yes

4. Algorithm

45. Initialize H to the most specific hypothesis (Ø in every attribute).
46. Read each training example from left to right.
47. If the target is No, ignore the example.
48. If the target is Yes, compare each attribute with H.
49. Replace Ø with the example value; if a value conflicts with the current specific value, replace it with ?; otherwise keep it unchanged.
50. Display the final hypothesis after all examples have been processed.

5. Step-by-step Procedure and Detailed Calculations

51. Start: $H_0 = (Ø, Ø, Ø)$.
52. Example 1 is positive: (High, Fast, Large). H_1 = (High, Fast, Large).
53. Example 2 is positive: (High, Fast, Small). Memory differs (Large ≠ Small), so H_2 = (High, Fast, ?).
54. Example 3 is negative (No), so it is ignored. H_3 = (High, Fast, ?).
55. Example 4 is positive: (Medium, Fast, Large). Price differs (High ≠ Medium), so Price becomes ?. Speed remains Fast and Memory is already ?. H_4 = (?, Fast, ?).

6. Formulae

Let $H = (h_1, h_2, \ldots, h_n)$ be the current hypothesis and $x = (x_1, x_2, \ldots, x_n)$ be a positive training example.

Attribute update:

- If $h_i = Ø$, then $h_i(new) = x_i$.
- If $h_i = x_i$, then $h_i(new) = h_i$ (no change).
- If $h_i \neq x_i$ and $h_i \neq Ø$, then $h_i(new) = ?$.

Final hypothesis for this dataset: Hfinal = (?, Fast, ?).

No numerical arithmetic is required for FIND-S: the “calculation” is a stepwise comparison of categorical attribute values. A mismatch causes generalization to ?.

7. Python Program

```
data = [
    ['High', 'Fast', 'Large', 'Yes'],
    ['High', 'Fast', 'Small', 'Yes'],
```

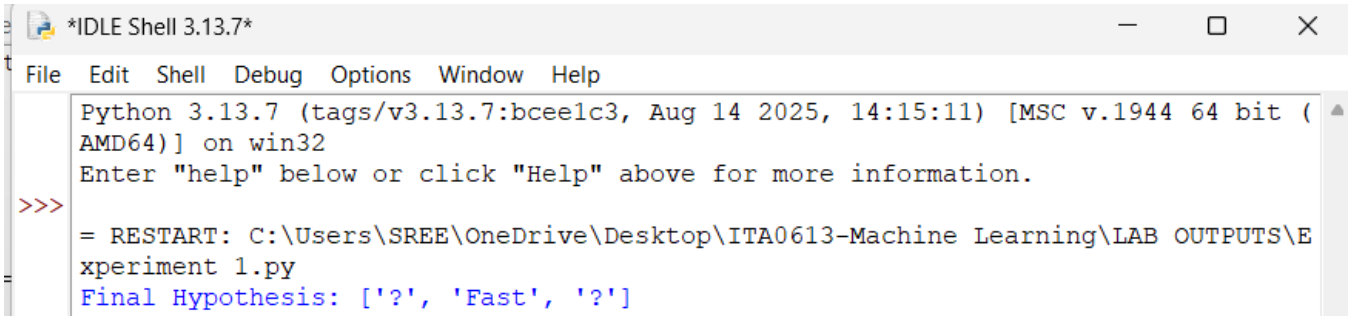
```
    ['Low', 'Slow', 'Small', 'No'],
    ['Medium', 'Fast', 'Large', 'Yes']
]

h = ['Ø'] * 3

for row in data:
    if row[-1] == 'Yes':
        for i in range(3):
            if h[i] == 'Ø':
                h[i] = row[i]
            elif h[i] != row[i]:
                h[i] = '?'

print("Final Hypothesis:", h)
```

8. Output



9. Result / Conclusion

The FIND-S algorithm was implemented successfully for this dataset. Final hypothesis: (?, Fast, ?). The positive examples share Fast speed, while Price and Memory vary.

Overall Result of Experiment 1

FIND-S was demonstrated on five datasets. The final hypotheses are summarized below.

Subproblem	Application	Final hypothesis
1	Enjoy Sport Prediction	(Sunny, Warm, ?, ?)
2	Weather Suitability Prediction	(Sunny, ?, ?, ?)
3	Student Performance Prediction	(?, Good, ?)
4	Animal Classification	(Warm, Four, ?)
5	Computer Purchase Prediction	(?, Fast, ?)

Final conclusion: FIND-S begins with the most specific hypothesis and generalizes only when positive examples require it. It ignores negative examples, so the learned hypothesis is not guaranteed to be the only possible concept consistent with the full dataset.

Notation: Ø = no value assigned yet; ? = any value. The exact output is based on the datasets listed in this record.